

**INSTITUTE OF ACTUARIES OF INDIA**  
**EXAMINATIONS**

**May 2026**

**SP2 - Life Insurance**

**Time allowed: 3 Hours 15 Minutes**

**Total Marks: 100**

**Q.1)** ABC Life has historically focused on agency-sold savings and protection products. Over the past three years, digital-first insurers have captured significant market share in urban term assurance, particularly among younger salaried customers. In response, the Board of ABC Life has approved development of a subscription-based digital protection product integrated with a fintech platform providing payment services and wearable health tracking.

### **Proposed Product Features**

- Yearly renewable term (YRT) assurance, with optional critical illness rider. Herein, the coverage is guaranteed for a year, and on renewal, the premium can be reviewed every year.
- Annual premium reset based on a proprietary health score derived from wearable data
- Simplified underwriting up to ₹40 lakh sum assured
- Premiums may be paused for up to three consecutive months without policy termination (No coverage during the premium pause period)
- Fully digital onboarding and servicing via API integration

Free surplus of the company is modest following recent growth. Solvency ratio stands at 161% against regulatory minimum of 150%.

i) The product is launched. After a year:

- Sales exceed projections by 35%
- Premium pause utilisation is materially higher (12%) than assumed (5%)
- Wearable engagement drops significantly after the first policy year. It stands at 52% relative to priced 80%.
- The reinsurer indicates that reinsurance rates may increase at next renewal

Explain the mechanisms through which these factors affect solvency, Embedded Value (EV) and profit emergence. [8]

ii) The Chief Financial Officer (CFO) argues that because the product is renewable and repriced annually, underwriting risk is limited and retention levels should be increased to improve margins. The Head of Underwriting disagrees, stating that behavioural effects and data asymmetry may increase long-term risk.

Highlighting the underwriting risks inherent to the product structure, evaluate the arguments made by the CFO and the Head of Underwriting. [6]

iii) Separately, the Chief Technology Officer (CTO) informs the Board that the wearable scoring algorithm is **not fully explainable** and the data flows depend **on third-party APIs**. Further, the premium reset process **has not been stress-tested** under high transaction volumes.

The regulator requests clarification on fairness, transparency and governance safeguards.

In light of the CTO's observations and the regulator's concerns, discuss the operational, and regulatory risks associated with the use of wearable-based algorithms, third-party APIs and automated premium reset processes, and their potential implications for the company. [6]

iv) Compare stochastic and deterministic modelling approaches and explain why stochastic modelling may be preferred for assessing future profits and capital risk for this product/in general. [4]

- v) ABC Life determines its solvency capital requirement using a Value-at-Risk (VaR) framework over a one-year horizon.

Define the key principles of a VaR-based solvency capital framework for life insurers, including the role of stress scenarios and diversification. **[3]**

- vi) Finally, the Board asks whether the product design could be modified to improve resilience without materially reducing growth ambition.

Suggest possible modifications to the product design that could improve capital stability, and/or risk management. **[3]**

**[30 marks]**

**Q.2)** ABC Life has experienced strong growth in its protection portfolio, particularly term assurance products sold through digital distribution channels. Over the past few years, the actuarial team has conducted a review of emerging experience. Key observations from recent experience analysis:

- Mortality experience has been higher than pricing assumptions for policies issued through certain digital channels.
- Early claim incidence (within the first policy year) has increased compared to previous cohorts.
- Persistency has been weaker than expected, particularly after the second policy anniversary.
- Several claim investigations have revealed non-disclosure of pre-existing medical conditions.

The Chief Actuary believes that some of these trends may reflect changes in customer demographics and underwriting practices following rapid growth.

At the same time, the company's statutory valuation date is approaching, and the appointed actuary must determine whether changes to valuation assumptions are required.

- i) Explain the steps that should be taken when conducting an experience investigation into the mortality experience of the protection portfolio. **[7]**

- ii) Discuss how the key observations from recent experience analysis described above may affect the assumptions used in the statutory valuation. **[7]**

**[14 marks]**

**Q.3)** ABC Life sells a range of life insurance products including traditional participating endowment policies, unit-linked savings policies, and term assurance with return-of-premium (ROP) features.

In recent months, the company has received a significant increase in customer complaints and surrender requests, particularly for policies sold through certain distribution channels.

Several policyholders have complained that:

- The surrender value in early policy years is significantly lower than premiums paid, and
- They were not adequately informed about surrender penalties at the point of sale.

The regulator has also issued a circular reminding insurers that policyholders should be treated fairly, and that surrender values should not result in unreasonable outcomes for customers.

ABC Life's management is therefore reviewing its surrender value framework across different products.

- i) Explain how surrender values would typically be determined for the following products, and why the resulting surrender values may differ between them:
- Participating endowment policies
  - Unit-linked insurance policies
  - Term assurance with return-of-premium features [7]
- ii) Explain why surrender values in early policy years may be significantly lower than premiums paid, and discuss how insurers should maintain fairness when setting surrender values for initial policy duration. [6]
- iii) Assess how mis-selling and poor customer communication may lead to increased surrender rates and customer complaints. [4]
- iv) Explain why regulators may place restrictions on surrender charges in life insurance products. [2]
- [19 marks]**

**Q.4)** ABC Life sells a range of long-term savings and retirement products including participating endowment policies, and deferred annuity products. Many deferred annuity products also include guaranteed minimum annuity rates (GMAR) and allow flexible vesting.

- i) Discuss how the nature of liabilities of these products should influence the investment strategy of ABC Life. [7]
- ii) Explain actions the insurer could take to manage or mitigate the ALM risks arising from these products. [7]
- iii) Over the past year, interest rates have risen significantly. Following this, ABC Life has observed increased surrender activity in participating policies.
- Explain the key asset–liability management risks arising from the changes in customer behavior described above. [4]
- [18 marks]**

**Q.5)** Following a major geopolitical conflict outside India, global financial markets have become volatile, with equity markets falling sharply and interest rates rising. ABC Life, selling a wide range of life insurance products including unit-linked plans, has observed significant switching by policyholders from equity funds to debt-oriented funds.

- i) Analyse how such geopolitical events may impact life insurance companies. [6]
- ii) Discuss the concerns for the insurer arising from significant fund switching in unit-linked policies. [4]
- [10 marks]**

**Q.6)**

- i) Specify the requirements of data that can be used for monitoring the emerging experience in a life insurance company. [4]

- ii) The actual expenses of the insurance company were higher than expected. Describe with examples how an insurance company can manage and control its expenses. **[5]**  
**[9 marks]**

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